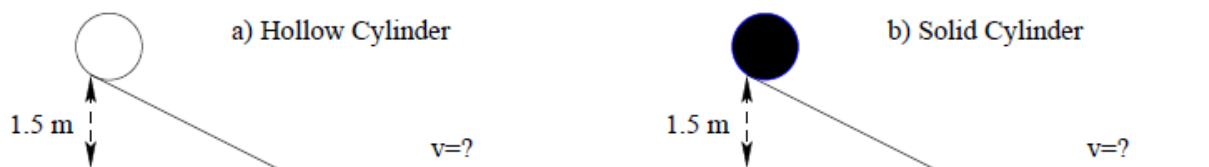
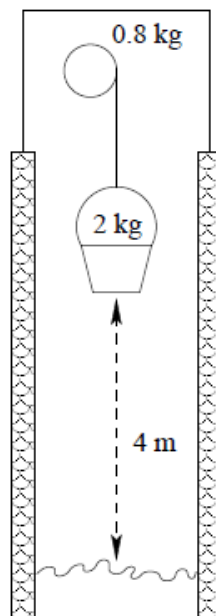


A race between two cylinders is performed. Both cylinders begin at the top of the 1.5 m tall ramp and roll without slipping to the bottom.



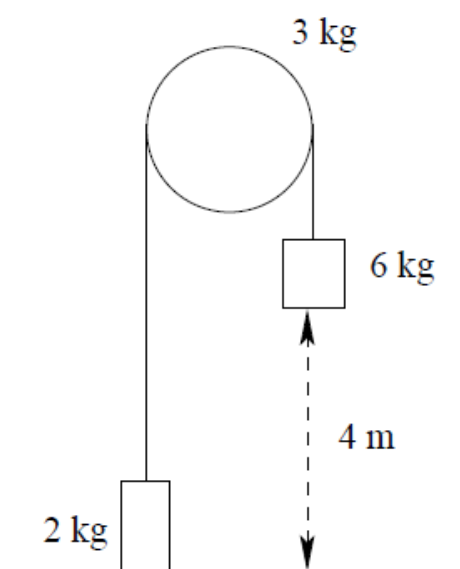
- (a) What is the speed of the hollow cylinder when it reaches the flat?
- (b) What is the speed of the solid cylinder when it reaches the flat?
- (c) Which one would win the race?

You are lowering a bucket into a well. When the bucket is 4 m above the water, your hand slips and the bucket falls toward the water while unwinding the spindle.



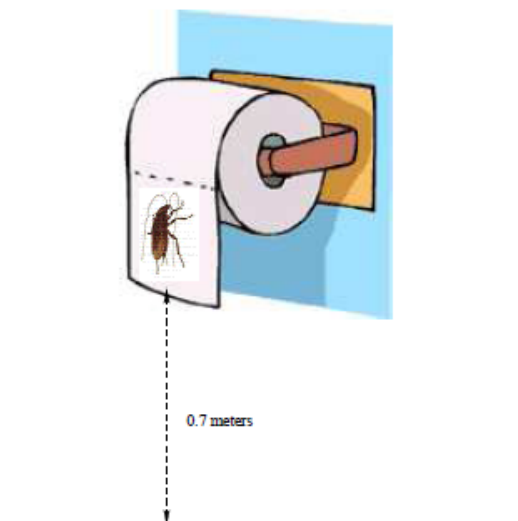
- (a) What is the speed of the bucket when it reaches the water?

Shown in the figure below is Atwood's machine. The masses are as labeled in the figure. Approximate the pulley as a solid cylinder.



(20 pts) Find the velocity of the 6 kg mass just before it strikes the floor.

Shown in the figure below is a scene from a filthy bathroom. As the roach clings to the paper, its weight causes the paper to unwind and the roach to move toward the floor. Treat the roll of paper as a solid cylinder of mass $M = 0.2 \text{ kg}$ and radius $r = 0.15 \text{ m}$. Treat the roach as a point object of mass $m = 0.1 \text{ kg}$.



- (a) (5 pts) Find the moment of inertia, I , of the roll of paper.
- (b) (15 pts) Find the velocity of the roach just before it hits the floor.